

(REVISION-2021)

MODEL QUESTION PAPER-1

THIRD SEMESTER--INSTRUMENTATION ENGINEERING

SUBJECT-ANALOG CIRCUITS FOR INSTRUMENTATION

[Time: 3 hours]

(Maximum marks: 100)

I: Part – A: Answer all the following questions in one word or sentence. (9 x 1 = 9 Marks)

1. Define current gain β .
2. Voltage gain of emitter follower configuration is ____
3. State the condition for Barkhausen criteria.
4. Phase shift in the feedback circuit of a RC phase shift oscillator with three stages.
5. Write one ideal Op-amp characteristic.
6. Define slew rate
7. State the use of precision rectifier
8. Write the name of any one comparator circuit
9. Write one application of clipper circuit.

II: Part - B: Answer any eight questions. Each question carries 3 marks (8 x 3 = 24 Marks)

1. Describe potential divide biasing for a common emitter amplifier?
2. Illustrate DC loadline.
3. Describe any three effects of negative feedback
4. Explain the feedback circuit of Wien bridge oscillator
5. Write a note on input impedance of Op- amp
6. Draw the circuit and state the gain for an inverting amplifier
7. Explain clippers using Op-amp
8. Illustrate zero crossing detector.
9. Write a note on log amplifier using op-amp
10. Describe common base configuration

Part - C: Answer all questions. Each question carries 7 marks.

(6 x 7 = 42 Marks)

III. Explain the frequency response of a common emitter amplifier.

OR

IV Explain the working of Common emitter BJT amplifier

V Explain the working of astable multivibrator using BJT.

OR

VI Explain the working of RC Phase shift oscillator.

VII Describe the block diagram of operational amplifier.

OR

VIII Design a non inverting amplifier to amplify a 5mV to 5V diagram.

IX. Explain sample and hold circuit

OR

X Explain the working of precision rectifier.

XI Distinguish the three transistor configurations.

OR

XII Explain the working of emitter follower.

XIII Explain any one LC oscillator circuit

OR

XIV Explain crystal oscillator.

Prepared by:

Nighila Paul
Lecturer in Inst. Engg.
GPTC koratty

Scrutinized by:

Chairman:
Designation:
Institution:

Question Wise Analysis

Course: **ANALOG CIRCUITS FOR INSTRUMENTATION –QP-1**

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.02	R	1	2
I.2	M1.07	R	1	2
I.3	M2.02	U	1	2
I.4	M2.03	U	1	2
I.5	M3.02	R	1	2
I.6	M3.03	R	1	2
I.7	M4.01	U	1	2
I.8	M4.02	R	1	2
I.9	M4.03	R	1	2
II.1	M1.02	U	3	8
II.2	M1.05	U	3	8
II.3	M2.03	U	3	8
II.4	M2.04	U	3	8
II.5	M3.03	U	3	8
II.6	M3.05	U	3	8
II.7	M4.03	U	3	8
II.8	M4.02	U	3	8
II.9	M4.05	U	3	8
II.10	M1.01	U	3	8
III	M1.06	U	7	16
IV	M1.04	U	7	16
V	M2.04	U	7	16
VI	M2.03	U	7	16
VII	M3.01	U	7	16
VIII	M3.05	A	7	16
IX	M4.04	U	7	16
X	M4.01	U	7	16
XI	M1.01	U	7	16
XII	M1.07	U	7	16
XIII	M2.03	U	7	16
XIV	M2.03	U	7	16

Course : ANALOG CIRCUITS FOR INSTRUMENTATION –QP-1

Blue Print

Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	15	31 ± 6	2	2	3	9	4	28	9	39
2	16	33 ± 6	2	2	2	6	4	28	7	36
3	14	29 ± 6	2	2	2	6	2	14	9	22
4	15	31 ± 6	3	3	3	9	2	14	6	26
Total	60	123	9	9	10	30	12	84	33	123

Blue Print

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	6	4.9
Understanding	110	89.4
Applying	7	5.7
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100

(REVISION-2021)

MODEL QUESTION PAPER 2

THIRD SEMESTER INSTRUMENTATION ENGINEERING

SUBJECT-ANALOG CIRCUITS FOR INSTRUMENTATION

[Time: 3 hours]

(Maximum marks: 100)

I: Part – A: Answer all the following questions in one word or sentence. (9 x 1 = 9 Marks)

10. Define bandwidth of a transistor amplifier
11. Output Characteristics of CE is plotted between _____ and _____.
12. Name any one LC oscillator circuit.
13. Phase shift in the feedback circuit of Wein bridge oscillator is _____.
14. Write the gain for inverting amplifier using Op-amp.
15. Input stage in the internal block diagram of the Op-amp is _____
16. Write one application of Schmitt trigger
17. State the use of precision rectifier.
18. Write one application of sample and hold circuit.

II: Part - B: Answer any eight questions. Each question carries 3 marks. (8 x 3 = 24 Marks)

11. Describe common base configuration
12. Illustrate input characteristics of a CE amplifier.
13. Draw the circuit for crystal oscillator
14. Describe the formula for the frequency of oscillation of astable multivibrator
15. Write notes on CMRR
16. Draw the circuit of an integrator using op-amp
17. Describe log amplifier
18. Write three uses of Schmitt trigger
19. Draw the circuit diagram of sample and hold circuit.
20. Describe differentiator using Op-amp.

Part - C: Answer all questions. Each question carries 7 marks.

(6 x 7 = 42 Marks)

III Explain the working of emitter follower

OR

IV Explain the working of Common emitter BJT amplifier

V Explain the working of Wein bridge oscillator

OR

- VI Explain any one LC oscillator.
- VII Describe the block diagram of operational amplifier.

OR

- VIII Design an inverting amplifier to amplify a voltage of 1mV to 1V
- IX. Explain Schmitt trigger circuit

OR

- X Explain the working of precision rectifier.
- XI Illustrate AC load line of a common emitter BJT amplifier.

OR

- XII Distinguish the three transistor configurations.
- XIII Explain the Barkhausen criteria with an example

OR

- XIV Explain the working of astable multivibrator

Prepared by:

Nighila V Paul
Lecturer - Inst. Engg.
GPTC Koratty

Scrutinized by:

Chairman:
Designation:
Institution:

Question Wise Analysis

Course: **ANALOG CIRCUITS FOR INSTRUMENTATION –QP-2**

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.06	U	1	2
I.2	M1.05	R	1	2
I.3	M2.03	R	1	2
I.4	M2.03	R	1	2
I.5	M3.03	R	1	2
I.6	M3.01	R	1	2
I.7	M4.02	R	1	2
I.8	M4.01	U	1	2
I.9	M4.03	U	1	2
II.1	M1.02	U	3	8
II.2	M1.05	U	3	8
II.3	M2.03	U	3	8
II.4	M2.04	U	3	8
II.5	M3.03	U	3	8
II.6	M3.05	U	3	8
II.7	M4.05	U	3	8
II.8	M4.02	U	3	8
II.9	M4.04	U	3	8
II.10	M3.05	U	3	8
III	M1.07	U	7	16
IV	M1.04	U	7	16
V	M2.03	U	7	16
VI	M2.03	U	7	16
VII	M3.01	U	7	16
VIII	M3.04	A	7	16
IX	M4.02	U	7	16
X	M4.01	U	7	16
XI	M1.05	U	7	16
XII	M1.02	U	7	16
XIII	M2.02	U	7	16
XIV	M2.04	U	7	16

Course : ANALOG CIRCUITS FOR INSTRUMENTATION –QP 2

Blue Print

Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	15	31 ± 6	2	2	2	6	4	28	8	36
2	16	33 ± 6	2	2	2	6	4	28	9	36
3	14	29 ± 6	2	2	3	9	2	14	8	25
4	15	31 ± 6	3	3	3	9	2	14	6	26
Total	60	123	9	9	10	30	12	84	31	123

Blue Print

Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	6	4.9
Understanding	110	89.4
Applying	7	5.7
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100